

Design and Development of low-cost bio inspired soft robot for pipeline inspection

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ABSTRACT---Pipeline inspection is vital for maintaining the integrity of oil, gas, and utility infrastructure. However, traditional inspection robots are often rigid, metallic, and unsuitable for navigating small-diameter or curved pipelines. This paper presents the design and development of a low-cost, bio-inspired soft robot crawler that utilizes pneumatic soft actuators for adaptive pipeline navigation. The crawler consists of two radial actuators and a central linear actuator made from hyperelastic silicone material, enabling inchworm-like motion. The components were modeled using SolidWorks and evaluated using ANSYS simulations to analyze deformation, strain, and stress under pressurized conditions. Molds were fabricated using 3D printing techniques and cast with Ecoflex™ 00-10 silicone to produce the soft actuators. This work contributes toward developing safer, more flexible, and scalable solutions for pipeline inspection in constrained environments.

I. INTRODUCTION

Pipeline systems are critical to energy, water, and utility infrastructures, transporting essential fluids across long distances. Ensuring the integrity of these pipelines requires regular internal inspections to detect blockages, corrosion, or structural damage. However, most existing pipeline inspection tools rely on rigid-bodied robots or manual procedures that are often unsafe, costly, and incompatible with small or curved pipe sections.

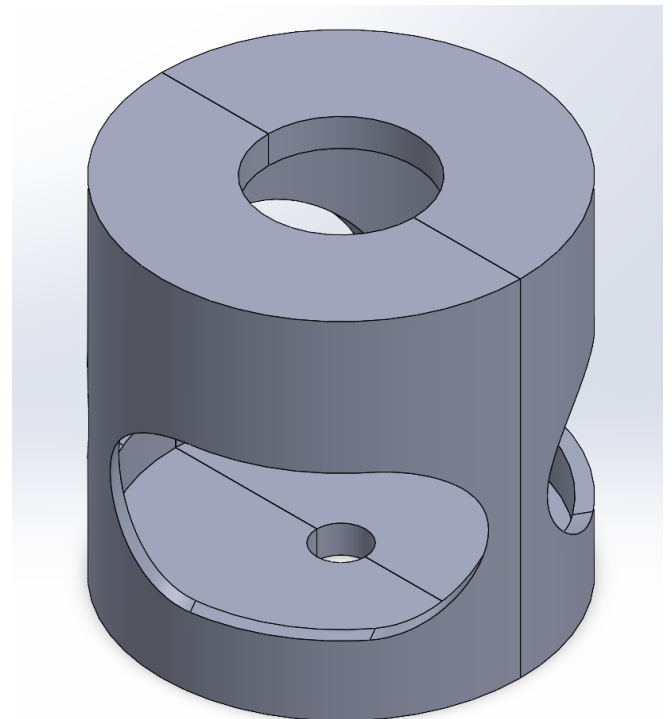
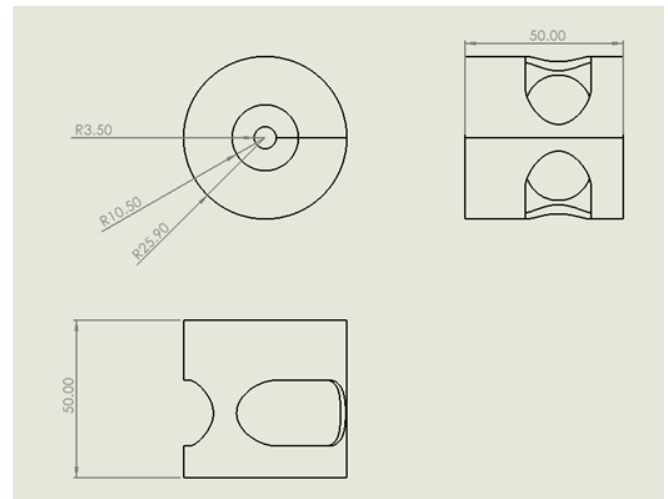
In industrial applications traditional inspection robots struggle to adapt to pipelines with diameters as small as 2 inches or environments with sharp bends. Additionally, the deployment of metallic, motor-driven crawlers can risk damaging pipe walls and may require large access points. Manual inspections, while still practiced in some cases, pose serious health and safety concerns. Companies often bear significant financial burdens due to insurance liability in the event of injury or equipment failure during inspections. Recent advances in soft robotics have introduced new opportunities for developing safer and more adaptable pipeline crawlers. Soft actuators, made from elastomeric materials, offer improved flexibility, safety, and conformability to irregular surfaces. Pneumatically driven soft robots have been explored in literature for bio-inspired motion systems that replicate inchworm-like locomotion, enabling movement through constrained geometries without the need for rigid structures or complex joints.

This paper presents the design and development of a soft robot crawler inspired by biological peristaltic motion, integrating two radial actuators and a central linear actuator fabricated using DOYB AND Ecoflex™ 00-10 silicone. The design aims to offer a low-cost, safe, and modular solution for non-invasive pipeline inspection, particularly for small and curved pipes.

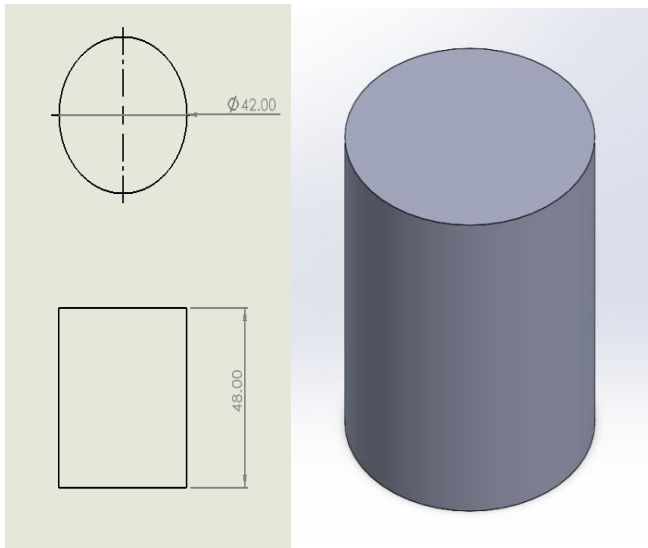
II. METHODOLOGY

A. Radial components (Shell and Balloon)

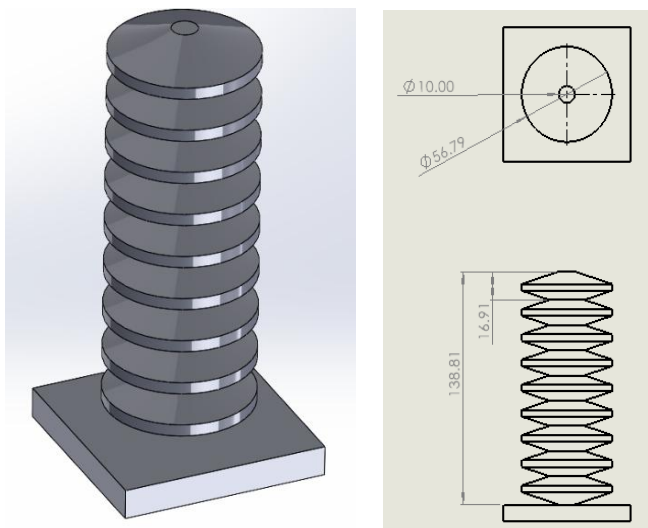
The radial actuator consists of two integrated components: a rigid outer shell and an internal expandable balloon. The shell was designed in SolidWorks with an outer diameter of 52 mm, a wall thickness of 4 mm, and a length of 50 mm. It features evenly distributed grooves created using the Wrap feature, which guide the expansion of the internal balloon outward during pressurization.



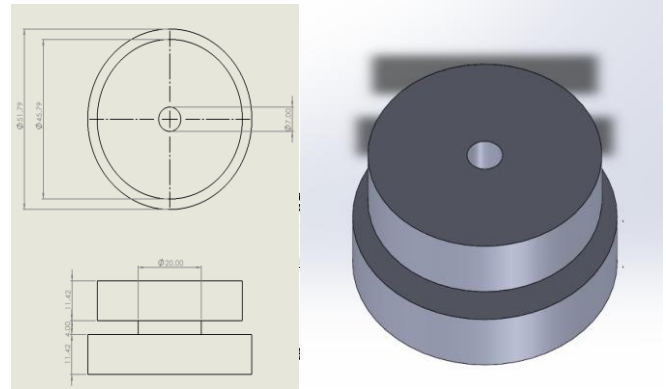
B. The balloon was modeled with a 48 mm diameter and 42 mm height, including a 5 mm central air inlet. The mold was a two-part cylindrical design, 3D printed in PLA using a Bambu Lab X1 Carbon printer, with a 36 mm core insert to ensure uniform 4 mm wall thickness. Core length was reduced by 4 mm to fully seal one end of the balloon during casting. The soft robot crawler was designed to mimic inchworm-like locomotion using a modular actuator system composed of two radial expansion units and one central linear actuator. Each actuator was modeled using SolidWorks 2023 and optimized for compatibility with small-diameter pipelines ranging from 50 mm to 75 mm. The radial units enable pipe wall anchoring, while the linear actuator provides axial propulsion. A rigid connector couples all components into a single functional body.



C. The linear actuator was modeled to produce axial elongation under internal air pressure, simulating linear propulsion. The final design measured 138.9 mm in length and included 9 bellows, each 16.91 mm long with an outer diameter of approximately 56.79 mm. The material used was **DoYB silicone**, a hyperelastic elastomer with properties similar to Ecoflex™ 00-10.

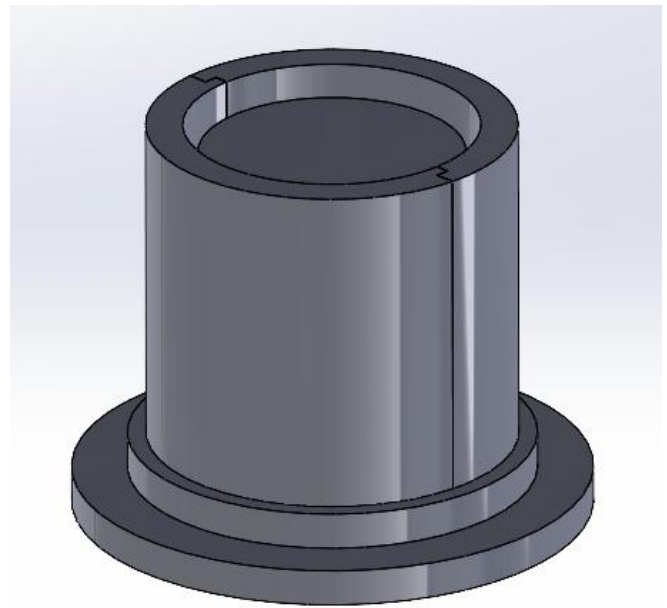


D. The connector was designed to mechanically and pneumatically interface the linear actuator and radial shell. It consists of two stepped cylinders: one 52 mm in diameter, matching the linear actuator end, and another 45.79 mm, designed to fit inside the radial shell. Both cylinders were 11.5 mm in length and separated by a 4 mm shoulder. A 5 mm air channel runs through the center, enabling air distribution. Abbreviations and Acronyms



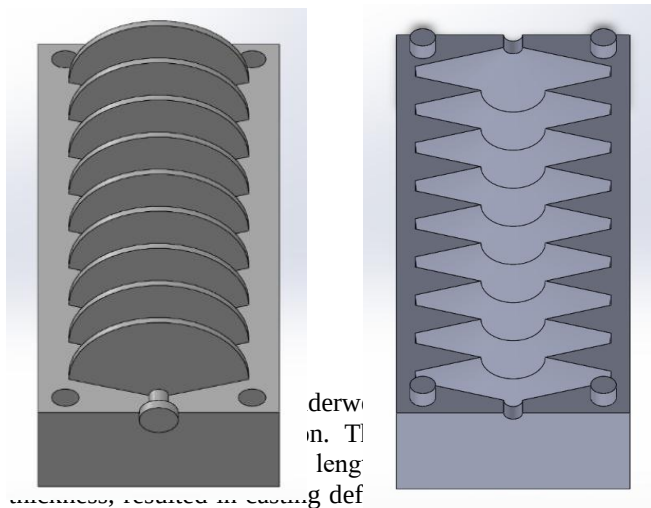
A. Radial Balloon Mold

The balloon mold was designed as a two-piece cylindrical system, with a 48 mm diameter outer cavity and a 36 mm centered core to achieve a 4 mm wall thickness. The mold was extruded to 46 mm in height, 4 mm longer than the balloon to allow full material sealing at one end. The mold was split vertically for easy removal of the cured part. A 5 mm channel was embedded to form the air inlet port.



B. Linear Actuator Mold

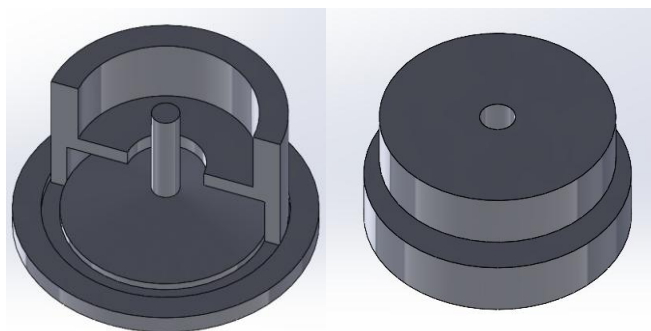
The linear actuator mold was created by halving the CAD model and embedding each side into rectangular PLA blocks using the Cavity feature. A 2 mm offset was applied to ensure wall thickness consistency. Alignment was maintained using integrated male and female pin features at the corners. This core-and-cavity configuration provided reliable shape accuracy and demolding without damage to delicate bellow geometries.



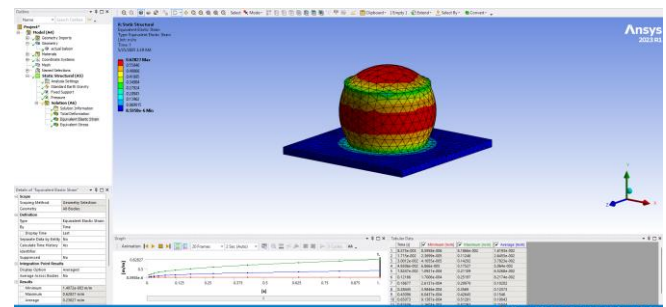
To overcome these issues, the final mold was redesigned with larger bellows ($\varnothing 56.79$ mm), fewer segments (9 bellows), and a 2 mm offset to improve structural integrity. The new design used a split-mold cavity approach with PLA blocks, supported by male/female alignment pins for precise sealing and easier demolding. These improvements led to an accurate actuator finish.

C. Connector Mold

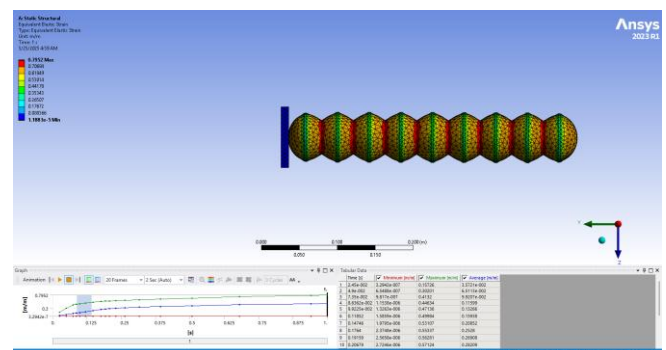
The final connector geometry itself was used as a solid reference and embedded directly into a circular mold block using the cavity feature. Similar to the balloon mold, the base included a 5 mm diameter circular core extruded upward by 20 mm to create the air inlet. This ensured the internal air channel remained aligned and open during pouring. The mold was designed for a single side pour and allowed for clean demolding due to the simplicity of the geometry and absence of flexible features.



The balloon was analyzed using a static structural study in ANSYS 2023 R1, focusing on its radial deformation behavior under pneumatic pressure. The CAD geometry was imported as a .STEP file and sectioned using a cut-plane to expose internal surfaces. These inner walls were defined using named selections, allowing pressure to be applied precisely across the inflatable area. The material was modeled using the Yeoh 3rd-order hyperelastic model, suitable for highly stretchable silicone like Ecoflex™ 00-10. The mesh used quadratic elements with an element size of 0.009 m to balance accuracy and solve time. A fixed support was applied to the base of the balloon, while 100 kPa of internal pressure was applied across the internal surface to simulate full inflation. Gravity was included along the $-Y$ axis, and large deflection was enabled to capture nonlinear expansion. The outputs analyzed were total deformation, elastic strain, and von Mises stress.



The linear actuator was subjected to a similar static structural setup, but with radial elongation as the primary deformation target. The actuator was also modeled in SolidWorks and imported into ANSYS as a .STEP file. A vertical base structure was included in the model to anchor the actuator and prevent unsupported displacement. Like the balloon, the Yeoh 3rd-order hyperelastic model was applied. The internal bellow surfaces were selected using named selections, and a pressure of 100 kPa was uniformly applied to these surfaces. The mesh used quadratic elements with a 0.009 m element size. A fixed support was applied to the rear base, and large deflection was enabled. This configuration allowed accurate analysis of axial deformation, elastic strain concentration, and von Mises stress in response to pressurization.



Ecoflex™ 00-10 was selected for its ultra-soft, stretchable characteristics suitable for radial balloon expansion. The material was modeled using the Yeoh 3rd-order hyperelastic model with supplier-based constants and assumed full incompressibility for simulation.

Property	Value
Material Name	Ecoflex™ 00-10
Density	1049 kg/m³
Hyperelastic Model Type	Yeoh – 3rd Order
Yeoh Constants (MPa)	C10 = 0.127, C20 = -0.149, C30 = 0.153
Source	From supplier data sheet
Incompressibility Parameters	D1 = 0, D2 = 0, D3 = 0
Assumption	Assumed fully incompressible

DoYB silicone was used as the linear actuator maaterial due to its comparable elasticity to Ecoflex. The same Yeoh 3rd-order model and material constants were applied, with assumptions based on similar commercial formulations and full incompressibility.

Property	Value
Material Name	DoYB Silicone
Density	1049 kg/m³
Hyperelastic Model Type	Yeoh – 3rd Order
Yeoh Constants (MPa)	C10 = 0.001, C20 = 0.0131, C30 = 0.00002056
Source	Approximated from similar Ecoflex model
Incompressibility Parameters	D1 = 0, D2 = 0, D3 = 0
Assumption	Assumed fully incompressible

III. RESULTS

Simulation analysis was essential to evaluate the structural behavior of the soft robot’s actuators under pneumatic loading. By applying static structural studies in ANSYS, key mechanical responses such as deformation, elastic strain, and stress distribution were assessed. This allowed validation of the actuator designs before fabrication, identifying potential failure zones and ensuring the materials operated within safe mechanical limits. The simulations also guided iterative improvements, ensuring that the components could achieve the required motion while maintaining structural integrity.

Balloon – Ecoflex™ 00-10

1. Total Deformation

Equation: $\Delta L = L_{\text{final}} - L_{\text{initial}}$

Where:

$L_{\text{initial}} = 49.79 \text{ mm}$ (from CAD)

$L_{\text{final}} \approx 91.38 \text{ mm}$ (from simulation)

$\Delta L = 91.38 \text{ mm} - 49.79 \text{ mm} = 41.59 \text{ mm}$

2. Elastic Strain

Equation: $\varepsilon = \Delta L / L_{\text{initial}}$

$\varepsilon = 41.59 \text{ mm} / 66.18 \text{ mm} \approx 0.628$ (62.8%)

3. Von Mises Stress

Stress value derived from ANSYS simulation to verify that material stress remains below the tensile limit. No manual equation required.

Linear Actuator – DoYB Silicone

1. Total Deformation

Equation: $\Delta L = L_{\text{final}} - L_{\text{initial}}$

$L_{\text{initial}} = 138.9 \text{ mm}$

$L_{\text{final}} \approx 381.09 \text{ mm}$

$\Delta L = 381.09 \text{ mm} - 138.9 \text{ mm} = 242.19 \text{ mm}$

2. Elongation (%)

Equation: $\text{Elongation (\%)} = (\Delta L / L_{\text{initial}}) \times 100$

$= (242.19 / 138.9) \times 100 \approx 174.36\%$

3. Elastic Strain

Equation: $\varepsilon = \Delta L / L_{\text{initial}}$

$\varepsilon = 242.19 \text{ mm} / 304.52 \text{ mm} \approx 0.7952$ (79.5%)

data summary tables for clear understanding

(Balloon)

Study	Result	Comments
Total Deformation	41.59 mm	Measured at dome center
Radial Expansion	Diameter: 91.38 mm	From original Ø49.79 mm to 91.38 mm
Elastic Strain	Max: 0.628 m/m, Avg: 0.238 m/m	Peak near top curvature
Von Mises Stress	85.4 kPa	Well below material limit (3.85 MPa)
Safety Factor	45	Highly safe, no rupture expected

(Linear Actuator)

Study	Result	Comments
Total Deformation	242.19 mm	Measured against the initial length
Elongation (%)	174.36%	From 138.9 mm to 381.09 mm
Elastic Strain	Max: 0.7952 m/m, Avg: 0.412 m/m	Peak at outer bellow crests
Von Mises Stress	15.818 MPa	Exceeds tensile strength (3.45 MPa)
Safety Factor	<1	Not safe, rupture risk

IV. DISCUSSION

The simulation and fabrication results of the soft robot crawler's actuators provided essential validation of their structural and functional behavior under internal pneumatic pressure. Each component was subjected to finite element analysis in ANSYS to predict deformation, strain, and stress distribution allowing assessment of safe operating and motion efficiency.

A. Balloon Actuator (Ecoflex™ 00-10)

The balloon actuator achieved a maximum radial deformation of 41.59 mm, expanding from a nominal outer diameter of 49.79 mm to approximately 91.38 mm. This expansion demonstrates sufficient wall-contact capability for anchoring within standard pipeline diameters ($\geq 50 \text{ mm}$). The maximum elastic strain reached 0.628 m/m, with an average of 0.238 m/m, indicating localized high-strain regions at the dome’s face consistent with its role in deforming outward. However, the strain remained well below the material's safe elastic limit (800% for Ecoflex 00-10), ensuring full elastic recovery upon depressurization. The von Mises stress peaked at 85.4 kPa, which is significantly below Ecoflex 00-10’s tensile strength (3.85 MPa), resulting in a safety factor of 45. This indicates robust resistance to failure, even under repeated inflation cycles. Such low-stress operation ensures long-term material stability and makes this actuator ideal for radial gripping tasks where uniform expansion and reusability are critical.

B. Linear Actuator (DoYB Silicone)

The linear actuator underwent a total elongation of 242.19 mm from an initial length of 138.9 mm, resulting in an elongation of 174.36%. This demonstrates strong linear displacement capabilities, mimicking earthworm-style motion critical for forward propulsion in confined pipes. However, this large deformation comes at a structural cost. The maximum elastic strain reached 0.7952 m/m, and average strain across the bellow segments was 0.412 m/m, with peak values concentrated at the outer crests of the bellows. This is near the upper threshold of safe elastic behavior for soft elastomers.

Most notably, the von Mises stress topped at 15.818 MPa, exceeding the tensile strength of DoYB silicone (3.45 MPa). This results in a safety factor <1 , indicating that the current design poses a high risk of rupture under full-pressure operation. These findings suggest that the actuator geometry needs further refinement potentially via wall thickening, reducing internal pressure, or switching to a stiffer material like Ecoflex™ 00-30. Geometry smoothing by adding more fillets could also be introduced to redistribute concentrated stress zones and minimize strain gradients.

C. Design Implications

The simulation results directly informed design decisions and mold adjustments. For the linear actuator, early simulation attempts failed due to mesh distortion and excessive strain in thin-walled bellows. By increasing the bellow diameter, decreasing their count, and offsetting mold geometry by 2 mm for uniform wall thickness, the revised actuator design achieved both manufacturability and performance targets, though the pressure and material limits remain a constraint.

In contrast, the balloon actuator's results validate both its design and material selection, confirming it as a safe and reusable anchoring unit within the robot. Its performance supports the radial gripping function necessary for stable locomotion.

V. CONCLUSION

This research aimed to design, simulate, and fabricate a low-cost, bio-inspired soft robot crawler capable of navigating confined pipelines using pneumatic soft actuators. Through detailed CAD modeling, mold design, material selection, and finite element simulation, the feasibility of using hyperelastic components for pipeline inspection was critically assessed.

The findings confirmed that while the radial balloon actuator performed safely and reliably within the elastic range of Ecoflex™ 00-10, the linear actuator experienced stress levels beyond the material's tensile strength. This supports the conclusion that soft robotic systems are a promising solution for pipeline inspection, but their performance is highly sensitive to actuator geometry and material choice.

Future work will focus on optimizing the linear actuator's structure and exploring alternative silicone formulations with higher stiffness or layered reinforcements. Additional studies will also aim to integrate sensors and control systems to enable self-directed motion and real-time inspection feedback.

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